INDUS UNIVERSITY

Mid-Semester Examination

Institute	IITE	**Exam Month / Year**	2025-2026
Program	B.Tech	**Branch**	CE/CSE/IT
Semester	IV	**Regular / Repeater**	Regular
Course Code	CE0418	**Course Name**	OPERATING SYSTEM
Date	30/03/2026	**Time**	01:00 PM to 02:30 PM
Day		**Maximum Marks**	40

Instructions :

Attempt all questions.
Make suitable assumptions wherever necessary. Figures to the right indicate full marks.

Q 1(A)	What is an Operating System? Explain any two types of operating systems.	**04**
(B)	Explain the process state transition with a neat diagram.	**04**
	OR	
(B)	A set of processes arrive at the CPU with the following burst times. Apply Non-Preemptive SJF scheduling and determine: Gantt Chart, Average Waiting Time and Average Turnaround Time	**04**

Process	Arrival Time	Burst Time
P1	0	7
P2	2	4
P3	4	2
P4	7	1

Q 2(A)	Define the Critical Section Problem. What are the three requirements for a solution?	**04**
(B)	What is a semaphore? Explain the different types of semaphores in detail with examples.	**06**
	OR	
(B)	Explain the producer-consumer problem and solve it using a semaphore. Write pseudo-code for the same.	**06**

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(C)	What is Deadlock? List the conditions that lead to deadlock. Explain the deadlock prevention techniques?	06
	OR	
(C)	Consider the snapshot of the system with Five Processes and four types of resources: A, B, C, D. The Currently Available set of resources is (1,5,2,0). Evaluate the content of the Need Matrix. Determine if the System is in a Safe State?	06

Processe	Allocated Resources				Max. Requirement			
	A	B	C	D	A	B	C	D
P0	0	0	1	2	0	0	1	2
P1	1	0	0	0	1	7	5	0
P2	1	3	5	4	2	3	5	6
P3	0	6	3	2	0	6	5	2
P4	0	0	1	4	0	6	5	2

Q 3(A)	What is fragmentation? Differentiate internal fragmentation and external fragmentation?	04
(B)	Examine the following allocation algorithms: 1) First-fit, 2) Best-fit, 3) Worst-fit	06
	OR	
(B)	1) Explain the Segmentation in detail with a suitable diagram. 2) Explain Swapping in Detail.	06
(C)	For the following page reference string: 3, 8, 2, 3, 9, 1, 6, 3, 8, 9, 3, 6, 2, 1, 3. (Solve using 3 Frames) 1) Find the page fault using page replacement policie First-In-First-out (FIFO)? 2) Find the page fault using page replacement policies Least Recently Used (LRU)?	06
	OR	
(C)	What is paging? Draw a neat and labelled diagram for the page table.	06

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